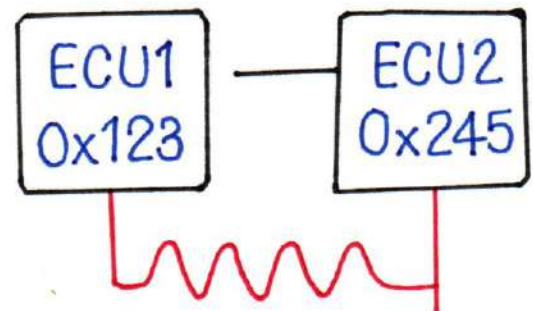
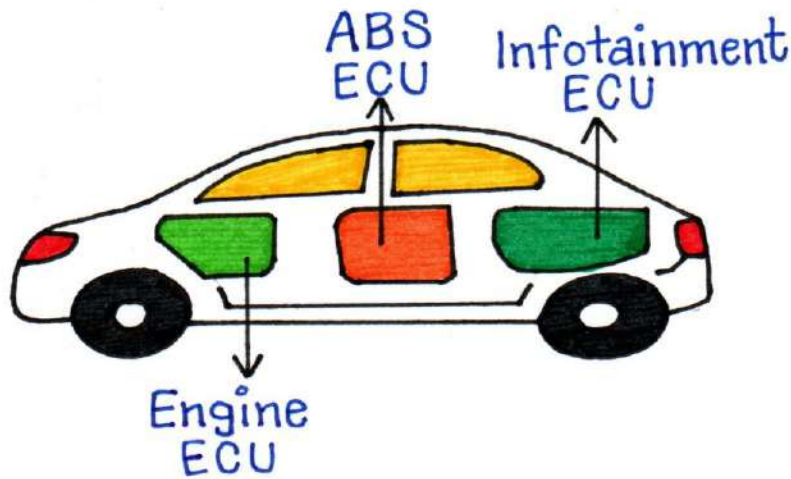


CAN PROTOCOL

The Communication Backbone
of Automobiles



Multiple ECUs communicate
over a single shared CAN Bus

Lower ID = Higher
Priority (Dominant
bits win)

SOF	Identifier	Payload	CRC + ACK	EOF
-----	------------	---------	-----------	-----

Each CAN message is a structured
frame ensuring priority & reliability.

Real-time torque reduction + brake indicator light -
all over CAN.

INTERVIEW GUIDE

Simplified • Handwritten • Interview Focused

CAN PROTOCOL

1. What is CAN Protocol ?

⇒ Controller Area Network (CAN) protocol is a standard for communication between various electronic devices like engine management systems, active suspension, ABS, Gear Control, lightning control, air conditioning, airbags, central locking, etc which is embedded in an automotive.

Features :-

- Speed
- Flexity
- Low Cost
- Robustness
- Built-in error detection

Applications :-

- Building automation
- Agricultural equipment
- Elevators, Escalators
- Medical instruments and equipment
- Industrial instrument, along with mechanical control
- Trucks, buses, passenger vehicles, gasoline vehicles & electric vehicles.

2. What is standard CAN and extended CAN ?

⇒ "CAN base frame" supports a length of 11 bits for the identifier which is made up of the 11-bits identifier whereas "CAN extended frame" supports a length of 29-bits for the identifier which is made up of an 18-bits extension.

3. What are CAN Messages ? List types ?

⇒ It is a message-based protocol, originally designed for multiplex electrical wiring within automobiles to save on copper.

The CAN messages types are listed below :-

- Data frame :- Data frame is a frame containing node data for transmission.
- Error frame :- Error frame is a frame transmitted by any node detecting an error.
- Remote frame :- A remote frame is a frame requesting the transmission of a specific identifier.
- Overload frame :- An overload frame is a frame to inject a delay between data or other remote frame.

4. Explain high speed CAN and low speed CAN ?

⇒ High speed CAN :- High speed CAN offers baudrate (from 40 Kbps to 1 Mbps) which is depending on the cable length. It has the most popular standard for the physical layer. It allows for simple cable connections between devices. It is the physical standard used in the DeviceNet and CANopen specifications.

Low-speed / Fault tolerant CAN :- It offers baudrates (from 40 Kbps to 125 Kbps). It is also a standard that allows CAN bus communication to continue in case of a wiring failure on the CAN bus lines. In the low speed / fault tolerant CAN network, each device has its own terminal termination.

5. What is SAE J1939 ?

⇒ SAE stands for Society of Automotive Engineers. SAE J1939 is the vehicle bus recommended standard basically used for communication and diagnostics among vehicle components.

6. What is ISO 11898 ?

⇒ ISO 11898 stands for the international standard organisation for high-speed CAN communication in road vehicles. It also specifies the PMA and MDA sublayers of the physical layers.

7. What is CAN open protocol ?

⇒ CAN open is a high-level communication protocol and also a device profile specification that is based on the CAN protocol. This protocol was developed for embedded networking protocol applications, such as in-vehicle networks. It also covers a network programming framework, device descriptions, interface definitions, and application profiles. CANopen provides a protocol that standardized communication between devices and applications from different manufacturers.

8. What is CAN and its uses ?

⇒ CAN is a multi-master broadcast serial bus standard for connecting ECU's.

- CAN or CAN-bus is a vehicle bus standard designed to allow microcontrollers or devices to communicate with each other within a vehicle without a host computer.
- CAN is a message-based protocol, designed specifically for automotive application but now also used in other areas such as industrial automation and medical equipment.
- The CAN bus is a serial asynchronous bus used in instrumentation applications for industries such as automotive.

Uses :-

- More reliable, e.g., few plug-in connectors that might cause errors.
- Wiring less complicated, more economic.
- Easy to implement, changes too.
- Additional elements (e.g., control units) are easy to integrate.
- Wire may be diagnosed.

9. How does CAN frame work ?

⇒ SOF - 1 Dominant

Arbitration field - 11 bit identifiers or 29-bit identifiers

Data byte - (0-8) bytes.

CRC field - 15 bits, Delimiter (1 bit recessive)

ACK field - 1 bit, Delimiter (1 bit recessive)

EOF - 7 bits recessive

IFS - 3 bits recessive.

Types of errors :- ACK error, Bit error, Stuff error, Frame error, CRC error.

10. Why CAN is having 120 ohms at each end ?

⇒ To minimize the reflection reference, to reduce noise. To ensure that reflection does not cause communication failures, the transmission line must be terminated.

11. Why CAN is message oriented protocols ?

⇒ CAN protocol is a message-based protocol, not an address-based protocol. This means that messages are not transmitted from one node to another node based on address. Embedded in the CAN message itself is the priority and the contents of the data being transmitted. All nodes in the bus system receive every message transmitted on the bus (and will acknowledge if the message was properly transmitted).

12. CAN logic what it follows ?

⇒ Wired AND logic.

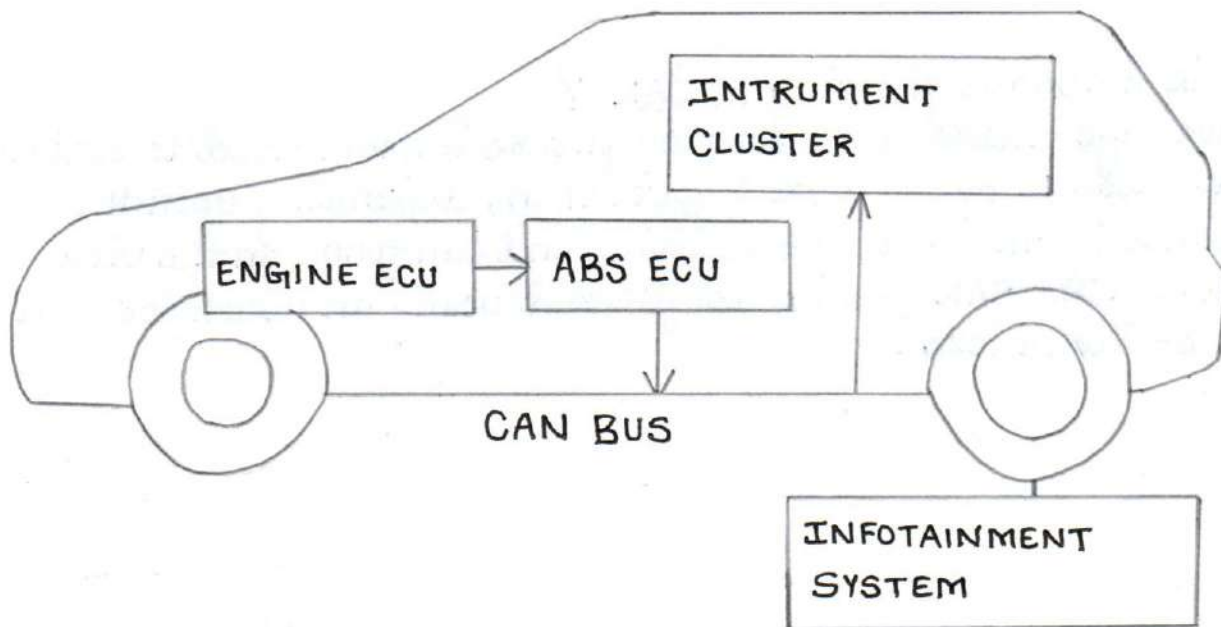
13. What is the speed of CAN ?

⇒ 40 m @ 1 Mbps and if the length of the cable increases will decrease the speed, due to RLC component on the cable.

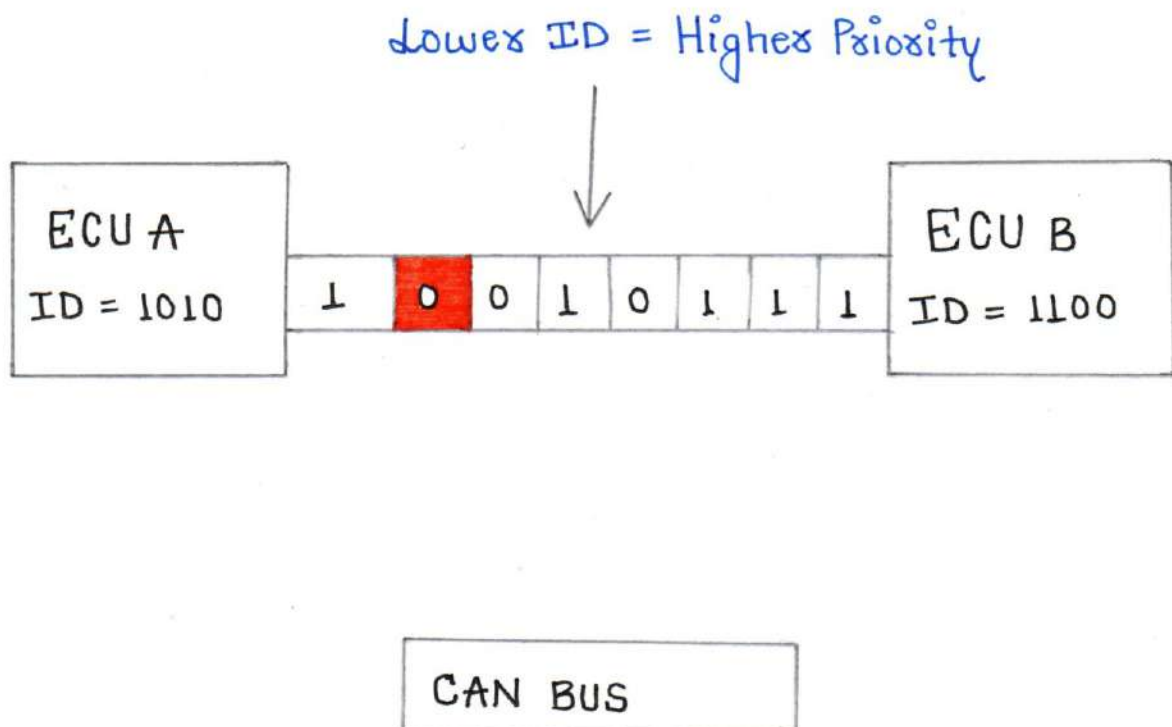
14. In CAN uni-cast and multi-cast also possible ?

⇒ We have to do small setting in ECUs for multi-cast and uni-cast, we have to do this setting in receiver node, this setting is called as message acceptance filter setting.

• Diagram 1 :- Automotive Use-Case



• Diagram 2 :- Arbitration Process



15. What is CAN arbitration and how CAN will follow the arbitration ?

⇒ CAN arbitration refers to the process in which two or more CAN controllers agree on who is to use the bus.

(or)

CAN arbitration is nothing but the node trying to take control on the CAN bus.

CSMA/CD :- Carrier Sense Multiple Access with collision Detection.

AMP :- Arbitration on Message Priority.

CSMA/CD + AMP



Two bus nodes have got transmission request. The bus access method is CSMA/CD + AMP. According to this algorithm both network nodes wait until the bus is free (carrier sense). In that case the bus is free both nodes transmit their dominant start bit (carrier sense multiple access). Every bus node transmits and reads back bit by bit from the bus during the complete message and compares the transmitted value with the received value.

As long as the bits are identical from both transmitters nothing happens. This method of arbitration will always cause the **lowest numerical value message ID** to have priority. This same method of bit-wise arbitration and prioritization applies to the 18-bit extension format as well.

16. If master sends 764 and slave sends 744 which will get the arbitration ?

⇒ Starts from MSB, first nibble is same, Master sends 7, slave also sends 7, the message with more dominant bits will gain the arbitration, lowest the message identifier higher the priority.

Example :-

• Device 1 :- address 443 (00110110001)

• Device 2 :- address 154 (00010011010)

• Device 3 :- address 187 (00010111011)

→ This device will take control to the bus among all the devices.

17. Standard CAN and Extended CAN difference ?

⇒ Number of identifiers can be accommodated for standard frame are 2^{11} i.e., 2^{11} .

Number of identifiers more compare to base frame, for extended frame are 2^{29} i.e., 2^{29} .

IDE frame bit - 1 for extended frame.

IDE bit - 0 for standard frame.

18. What is bit-stuffing ?

⇒ CAN uses a Non-return-to-zero protocol, NRZ-5 with bit stuffing. The idea behind bit-stuffing is to provides a guaranteed edge on the signal so the receiver can synchronise and resynchronise with the transmitter before minor clock discrepancies between the two nodes can cause a problem.

With NRZ-5 the transmitter transmits at most five consecutive bits with the same value. After five consecutive bits with the same value (zero or one), the transmitter inserts a stuff bit with the opposite state.

19. What is the use of bit stuffing ?

⇒ Long NRZ messages causes problems in receivers.

- Clock drift means that if there are no edges, receivers lose track of bits.
- Periodic changes edge allows receiver to resynchronize to sender clock.

20. Functionality of data link layer in CAN ?

⇒ LLC (Logical Link Control) :- Overload control, notification, Message filtering and recovery management functions.

MAC (Medium Access Control) :- Encapsulation/De-capsulation, error detection and control, stuffing and de-stuffing and serialization and de-serialization.

21. What is meant by synchronization ?

⇒ Synchronization is timekeeping which requires the co-ordination of events to operate a system in unison.

22. What is meant by hard synchronization and soft synchronization?

⇒ Hard Synchronization :- It is done with a falling edge on the bus while the bus is idle, which is interpreted as a start of frame (SOF). It starts and restarts the internal Bit Time Logic.

Soft Synchronization :- It is used to lengthen or shorten a bit time while a CAN frame is received.

23. What is the difference between function and physical addressing?

⇒ Functional addressing is an addressing scheme that labels messages based upon their operation code or contents.

Physical addressing is an addressing scheme that labels messages based upon the physical address location of their source and/or destination(s).

24. What happens if I have to send more than 8-bytes of data?

⇒ The J1939 standard has defined a method of communicating more than 8-bytes of data by sending the data in packets as specified in the Transport Protocol (TP). There are two types of TP, one for broadcasting the data and the other for sending it to a specific address.

DTC consists of 4 components :-

SPN - Suspect Parameter

FMI - Failure Mode Identifier

OC - Occurrence count

GM - Conversion method

25. What is OBDII?

⇒ On-Board Diagnostics is an automotive context is a generic term referring to a vehicle's self-diagnostic and reporting capability.

26. Why Diagnostics standard?

⇒ As system got more complex the link between cause and symptom becomes less obvious. This means that electronic system had to have some level of self diagnosis and to communicate to the outside world. Initially many systems used their own protocols which means that garages had to have a large number of tools - even to diagnose a single vehicle.

27. Can you have two transmitters using the exact same header file field?
⇒ No, that would produce a bus conflict.

- Unless you have middleware that ensure only one node can transmit at a time.
- For example use a low priority message as token to emulate token-passing.

28. What is CAN physical layer voltage level?

⇒ CAN High :- 2.5 to 3.5V

CAN Low :- 1.5 to 2.5V

Dominant bit :- 0

Recessive bit :- 1.

29. What happens when two nodes are sending same identifiers at a same time?

⇒ Two nodes on the network are not allowed to send messages with the same ID. If two nodes try to send a message with the same ID at the same time arbitration will not work. Instead, one of the transmitting one of the transmitting nodes will detect his message is distorted outside of the arbitration field. The nodes will then use the error handling of CAN, which in this case ultimately will result in one of the transmitting node being switched off (bus-off mode).

30. What is meant by verification and validation?

⇒ Verification & validation (V&V) is the process of checking that a software system meets specifications and it fulfils its intended purpose. It is normally part of the software testing process.

- Verification :- The process of evaluating software to determine whether the product of a given development phase satisfies the conditions imposed at the start of that phase.
(08)

Verification shows conformance with specification.

- Validation :- The process of evaluating software testing during or at the end of the development process to determine whether it specifies requirement.
(08)

Validation shows that the process of program meets the customer's needs .

31. What is the difference between bit rate and baud rate ?
⇒

Bit Rate

1. Bit rate is defined as the transmission of a number of bits per second .
2. Bit rate is also defined as per second travel number of bits .
3. Bit rate emphasizes computer efficiency .
4. Bit rate = baud rate $\times$ the number of bit per baud .
5. Bit rate not used to decide the requirement of bandwidth for transmission of the signal .
6. Bit rate cannot determine the bandwidth .
7. It counts the number of bits travelled per second such as Kbps , Mbps , Gbps , etc .

Baud Rate

1. Baud rate is defined as the number of signal units per second .
2. Baud rate is also defined as per second number of changes in the signal .
3. Baud rate emphasizes data transmission .
4. Baud rate = bit rate $\times$ the number of bit per baud .
5. Baud rate is used to decide the requirement of bandwidth for transmission of the signal .
6. Baud rate can determine the amount of bandwidth necessary to send the signal .
7. It counts how many times the state of a signal is changing .

32. What is meant by active and passive error mode ?

⇒ A node starts out in Error active mode .

When any one of the two Error counters raises above 127 , the node will enter a state known as Error Passive and when the transmit Error counters raises above 255 , the node will enter the Bus off state .

Active node will transmit Active Error flags when it detects errors .

33. What are the functions of CAN Transceivers ?

⇒ The transceiver provides differential transmit capability to the bus and differential receive capability to the CAN controller. Transceiver provides an advanced interface between the protocol controller and the physical bus in a CAN node.

Typically, each node in a CAN system must have a device to convert the digital signals suitable for transmission over the bus cabling (differential output). It also provides a buffer between the CAN controller and voltage spikes that can be generated on the bus CAN by outside sources.

34. Why CAN has given two wires for data ?

⇒ Reason, differential voltage method is less sensitive to external noise.

(Single voltage method is more sensitive to external noise and it can corrupt the data).

- It is a type of asynchronous serial communication protocol because no common clock is used between ECU's.

CAN Protocol is :-

- 1) Broadcast
 - 2) Multi-master (priority based bus access)
 - 3) Half Duplex
 - 4) Asynchronous serial communication protocol.
- Maximum node on CAN bus is particularly up to 30 nodes.

CAN working layer :-

- 1) Data-link layer (ISO 11898-1)
 - ↳ CAN controller
- 2) Physical layer (ISO 11898-2/3)
 - ↳ CAN Transceiver.

35. What is the process of the terminating resistor on CAN lines ?

⇒ A termination resistor is an electrical component that is used to match the impedance of a transmission line to that of the connected load. It is typically used at the end of a transmission line to prevent signal reflections that can degrade the quality of the transmitted signal.

36. What is message filtering in CAN ?

⇒ The message filters are just easy filters implemented to restrict the data logged by the data logger. Then, depending upon the importance of the data, the receiving end can focus on the vital messages. The CAN frames are filtered using an identifier or a range of identifiers.